Shifa Somji

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Research Interests

I am interested in pushing the boundaries of generative, agentic, and physical AI, full-stack robotics, and human-robot interaction, enabling robots to interact with the physical world safely, fluently, and naturally.

Education

Purdue University West Lafayette, IN

Ph.D. in Computer Science, Advisor: Sooyeon Jeong *Aug 2023 - present*

M.S. in Computer Science, GPA: 3.97 *Aug 2023 - May 2025*

Relevant Coursework: Deep Learning, Robot Learning, Computer Vision, Natural Language Processing, Software Engineering and Robotics, Human Robot Interaction, Statistical Machine Learning, Algorithms, Distributed Systems

Harvey Mudd College Claremont, CA

B.S. in Computer and Cognitive Science, GPA: 3.7 *Aug 2019 - May 2023*

Relevant Coursework: Artificial Intelligence, Machine Learning, Neural Networks, Computer Vision, Statistical Linear Models, Operating Systems, Databases, Managing Data at Scale, Algorithms, Programming Languages, Software Engineering, Computability and Logic, Cognitive Science, Linguistics

Skills

- AI Frameworks: PyTorch, TensorFlow, TensorRT, OpenClaw, LangChain, NVIDIA NeMo, NVIDIA Triton, NVIDIA Issac
- Programming Languages: Python, Java, C++, JavaScript, React, Racket, ROS, MATLAB, SQL
- Tools: Kubernetes, AWS, Workflows, Cursor, Codex, Spark, Jupyter, Git, Bash, Latex

Research Experience

Proactive Learning in Human-Robot Collaboration - CoMMA Lab May '25 - Present

- Advanced uncertainty-aware socially assistive robots that proactively identify knowledge gaps, generate contextually meaningful questions, and learn personalized user preferences across 100+ repeated interactions through long-term personalized adaptation.
- Implemented robust preference-learning algorithms that model noisy and inconsistent human corrections, enabling long-term memory retention, and reducing repeated instruction during human–robot collaboration.
- Established evaluation frameworks for long-term teachability, including interaction benchmarks, multimodal behavioral analysis, and novel metrics that quantify knowledge retention and human teaching burden over time.

Personalized Speech Therapy Robot for Aphasia Patients - HAI Lab Aug '23 - May '25

- Developed, trained, and personalized socially assistive robots (SARs) as a long-term speech therapy solution for patients with aphasia, a neurogenic post-stroke condition that results in communication deficits, language production, and comprehension. My research shows that SARs can deliver successful scalable speech language therapy and adapt their interaction style based on participants' unique needs and preferences in a home context.
- Incorporated large language models (LLMs) to generate real-time personalized responses and adjust SAR's interaction style based on user input.
- Employed efficient text to speech (TTS) systems, analyzed facial expressions and behavioral cues, and conducted data analysis of qualitative surveys.

Algorithmic Search Framework - AMISTAD Lab Jan '21 - May '22

- Proposed an algorithmic search framework, unifying machine learning and artificial intelligence under the single research discipline of artificial learning. The framework bounds the performance and provides an intuitive understanding of artificial learning algorithms.

xenoGI 3 - Bush Lab May '21 - Aug '21

- Developed a new simulator that evolves sequences over a user-provided phylogenetic tree to test xenoGI, a software package that identifies genomic islands, maps their origin within a clade of closely related bacteria while allowing for horizontal transfer of novel genes as well as for genomic scale deletions, duplications and inversions, and amino acid level sequence change. Such information is valuable because it helps us understand the adaptive path that has produced living species.
- Created a plotting feature for reconciliation between gene and species trees. This feature overlaps the species and gene trees and adds different markers for origin, speciation, rearrangement, duplication, and transfer, identifying the location of each event.

Publications

- **Somji, S.**, Kingston, Z. Proactive Teachability in Socially Assistive Robots through Uncertainty-Aware Clarification. (2026).
- Bui, T., **Somji, S.**, Merzdorf, H., Kong, N., Tay, L., Jeong, S. Beyond Surveys: Predicting Student Well-being from Open-ended Video Responses using LLMs. (2026).
- **Somji, S.**, Lee, J., Jeong, S. Personalization of Speech Therapy Treatment with Socially Assistive Robots. Toward Relationship-Centered Care with AI: Designing for Human Connections in Healthcare - CHI Conference on Human Factors in Computing Systems. (2026).
- **Somji, S.**, Gergerli, G., Lee, J., Jeong, S. Language Alignment with Socially Assistive Robots in Older Adults: Implications for Aphasia Rehabilitation. The 62nd Annual Meeting of Academy of Aphasia. (2024).
- Liu, N., Gonzalez, T.A., Fischer, J., Hong, C., Johnson, M., **Somji, S.**, Wirth, J., Libeskind-Hadas, R., Bush, E. xenoGI 3: using the DTLOR model to reconstruct the evolution of gene families in clades of microbes. BMC Bioinformatics 24, 295 (2023).

Industry Experience

PayPal San Jose, CA
Agentic Engineering PhD Intern *May '26 - Aug '26*

- Built an inference optimization engine for PayPal's e-commerce agent platform to accelerate large language model inference.
- Researched and implemented state-of-the-art inference acceleration techniques, including speculative decoding, KV caching, quantization, and batching, to optimize latency, throughput, and GPU utilization for production agentic AI workloads.
- Designed a serving-time-aware optimization algorithm that dynamically selects inference policies under latency and resource constraints, reducing agent response latency by 12%, cutting inference cost by 50%, and improving LLM-as-a-Judge evaluation scores from 0.7 to 0.9.

Impinj Seattle, WA
ML Platform Intern *June '25 - Sep '25*

- Built and evaluated an RFID Electronic Article Surveillance system on 100K+ tag transition events, enabling real-time detection of merchandise movement through retail security gates.
- Applied Shapley Theory across 8 RFID sensor features, identifying the dominant predictors of tag transitions and evaluating model robustness across changing retail traffic patterns.
- Optimized LSTM model architecture for real-time RFID tag detection, reducing detection latency by 40% and maintaining 95% detection accuracy for real-time deployment.

Impinj Seattle, WA
Platform Architecture Intern *May '23 - Aug '23*

- Primary engineer responsible for development of an RFID-based logistics system for a major shipping company, enabling automated package sorting, routing, and tracking across 10 distribution centers.
- Trained and benchmarked Gradient Boosting, Artificial Neural Network, and baseline models on thousands of package tracking events to optimize RFID system parameters, improving package identification accuracy from 75% to 95%.
- Developed analytics pipelines to identify misloaded and unexpected packages. Evaluated alternative system configurations to improve tracking reliability and operational efficiency.

Harvey Mudd Clinic x FedEx Claremont, CA

Engineering Manager *Aug '22 - May '23*

- Led a team of five students working on a Harvey Mudd senior capstone project with the FedEx Autonomous Vehicle Deliverability Team.
- Created a Python model to determine necessary factors for autonomous vehicle delivery and provided an interpretable decision when to prioritize AV delivery for specific packages, resulting in an estimated 800,000 labor hours saved annually from isolated and failed signature deliveries.

Meta Seattle, WA

SWE Intern *May '22 - Aug '22*

- Interned on the Curated Ads Data team, responsible for building the platform to enable transparency, control, and purpose in FB advertisements.
- Created a system that intelligently identified duplicate requests for curated ads, identifying 20% of curated requests as duplicates and significantly improving curation throughput.
- Authored a Dataswarm pipeline, using SQL and Python, that created and updated a datastore of hashed values of requested CDS columns. Integrated another Dataswarm pipeline, using Hack, in Curated Data Management System (CDMS) that checked new requests against the datastore of hashed values and notified the users of duplicate requests.

TeamTime Claremont, CA

Co-Founder/CTO *May '20 - Aug '20*

- Co-founded TeamTime, a startup where members of a team can set group alarms for better collaboration and enhanced productivity.
- Designed and developed the application's persistence layer, comprising of users, groups, and alarms, using Firestore, a NoSQL database with data sync support and instant change listeners.
- Developed the app's user interface using React Native Elements and React Native Material UI packages with custom CSS.

Awards

- NSF Graduate Research Fellowship, *National Science Foundation*
- Young Investigator Fellowship, *Academy of Aphasia*
- John W. Anderson Foundation Scholarship, *Purdue Regenstrief Center for Healthcare Engineering*
- Graduate Teaching Award, *Purdue Department of CS*
- Class of '94 Award, *HMC Department of CS*
- Departmental Honors, *HMC Department of CS*
- Dean Chris Sundberg Prize, *HMC*

Teaching Experience

Teaching Assistant, Purdue University

- Foundations of Computer Science (Spring 2025, Fall 2023)
- Human Computer Interaction (Spring 2024)

Teaching Assistant, Harvey Mudd College

- Programming Languages (Spring 2023, Fall 2022, Spring 2022)
- Software Engineering (Fall 2021)
- Data Structures and Program Development (Spring 2021)
- Principles of Computer Science (Fall 2020)

Leadership

Graduate Women in Science Program • Computer Science Representative Aug '25 - Present

- Planned and organized 8 annual professional development events attended by 70–100 graduate students, covering topics in career advancement, research success, leadership, and life skills for STEM researchers.

Center for Innovation & Entrepreneurship • Business Development Manager Aug '21 - May '23

- Supported entrepreneurship programming for 80+ students by organizing 6 annual networking and experiential learning events connecting students with founders, investors, and industry professionals.

Women of the Association of Computing Machinery • Co-Chair Aug '21 - May '22

- Led technical resume workshops and coordinated 4 annual company-sponsored networking events, engaging 50+ students per event and supporting the professional development of women in computing.

Washington Student Math Association • President Sep '16 - Jul '19

- Led a student leadership team representing 25 schools across Washington.
- Organized an annual premier math competition, Math Bowl, for 200+ middle school mathaletes.

Chess4Girls • CEO Sep '15 - Aug '19

- Hosted motivational workshops for all participating girls at Washington State Elementary Chess Championship.
- Ran a monthly chess strategy session for all girls in King County.
- Sponsored girl-exclusive chess tournaments in the State of Washington including Washington Girls Chess Championship and Queen's Quest Chess Tournament.